1.1 Rewriting Numbers in Equivalent Forms

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I can explore patterns in fractions whose decimal equivalent is repeating.</td></tr>
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<tr><td colspan="2">6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals and percentages.</td><td colspan="2">MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.</td></tr>
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What are repeating decimals and which fractions have decimal expansions that repeat?</td></tr>
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<tr><td colspan="2"><u>Key Terms:</u>
Rational number
Repeating decimal
<u>Additional Terms:</u>
Terminating
Ellipse</td><td>Career Profile Video
Scientific calculators</td><td>Preview the calculator displays students will be using in class to determine how repeating decimals will appear on screen
Warm-Up 1.1.1 contains a video to accompany the question prompts.</td></tr>
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<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not remember how to rewrite rational numbers in equivalent forms.
Students may forget to add the bar above the digits in a repeating decimal.
Students may get confused about repeating decimals if their calculator rounds the last digit of a repeating decimal.</td><td>Determine whether the scientific calculators that students will be using will round repeating decimals or not. If so, plan for how to address that with students so that they can interpret the technology accurately. Consider using the Desmos scientific calculator.
Working with probabilities in different forms throughout the remainder of this unit and then operations with rational numbers in the next unit will provide plenty of opportunities to scaffold and address any gaps revealed in this lesson, rather than stopping grade-level instruction to re-teach this content.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
For 1.1.5 Bring It Together, it is not expected that students come up with a rule that always works. However, students' conjectures will reveal helpful information about what patterns they are noticing in these relationships.
Consider implementing a Think-Pair-Share here and having students share what they notice and talking about these observations as a class after discussing with their partner.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Students explore which fractions have decimal expansions that repeat using a scientific calculator. They also make connections between equivalent fractions in this context. After exploring, students use the patterns they observe to make and refine predictions about which fractions result in repeating decimal expansions before making a conjecture at the end of the lesson. Support students to develop generalizations based on the similarities found among problems as they begin developing a sense of repeating decimals.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">As gaps in mastery are revealed, consider using the On-Ramp tool to address specific gaps and determine the best way to scaffold throughout the remainder of this unit. Working with probabilities in different forms throughout the remainder of this unit and then operations with rational numbers in the next unit will provide plenty of opportunities to scaffold and address these gaps rather than stopping grade-level instruction to re-teach this content.</td></tr>
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1.1.1 Warm-Up: Day in the Life of a Mushroom Farmer

Component Overview: *Students engage with a video about the life of a mushroom farmer. Students are asked about the concept of adapting in life to begin establishing a culture of growth mindset in the classroom.*

1.1.1 Warm-Up: Day in the Life of a Mushroom Farmer

1. Alejandro, a mushroom-kit entrepreneur, says that when it comes to growing his business, "The most important thing is … adapting opportunities that come to you and still holding the same amount of passion."

   a. What is something you are passionate about?
   Answers will vary. I am passionate about cooking and playing sports.

   b. How has a new interest or hobby surprised you?
   Answers will vary. A new interest that surprised me was creating Tik Tok videos because I am normally shy about dancing in front of people.

Play the Career Profile video on the mushroom farmer for the entire class. After students watch the video (3:46 minutes), prompt them to reflect on the profession by answering the two questions independently.

Consider beginning to incorporate language of growth mindset, taking initiative or risk, and other key characteristics of your classroom culture you want to build.

Identify which qualities are important for you and your students and use this video as a starting point. "How does Alejandro show __________?"

1.1.2 Refresher: Rewriting Rational Numbers in Different Equivalent Forms

Component Overview: *Students review how to rewrite rational numbers in different forms from the 6th grade accelerated course. This component includes multiple strategies students can use to rewrite rational numbers in equivalent forms.*

1.1.2 Refresher: Rewriting Rational Numbers in Different Equivalent Forms

Rational numbers can be expressed as fractions, decimals, or percentages. Probabilities can be expressed using these forms as well.

1. Andrés and Jolie are discussing how to write $\frac{17}{20}$ as a decimal.

   Andrés says he can use long division to divide 17 by 20 to get the decimal.

   Jolie says she can write an equivalent fraction with a denominator of 100 by multiplying by $\frac{5}{5}$, then writing the number of hundredths as a decimal.

   a. Explain whether one, both, or neither of these strategies would work.
   Answers will vary. Both of these strategies will work because these are some of the methods to use to correctly convert fractions into the equivalent decimals.

   b. Rewrite $\frac{17}{20}$ in its equivalent decimal form. Show your work.
   $\frac{17}{20} \cdot \frac{5}{5} = \frac{85}{100} = 0.85$

2. Complete the statement.

   A percentage is a ratio expressed as a fraction with a denominator of $\underline{100}$.

This Refresher is meant to review what students have learned in the 6th grade accelerated course. As students work with their partners to complete the activity, gaps in mastery may be revealed; consider using the On-Ramp tool to address specific issues and determine the best way to scaffold throughout the remainder of this unit.

3. There are many strategies that can be used to rewrite rational numbers in different forms. Some of them are included below.

   a. Work with your partner to match each phrase in the column on the left with a strategy in the column on the right. Draw arrows to make each match.

Rewriting Numbers in Different Forms	Strategy
To rewrite a fraction as a decimal, …	… find an equivalent fraction with a denominator of 100.
To rewrite a decimal as a fraction, …	… express it as a fraction out of 100 and simplify, if necessary.
To rewrite a fraction as a percentage, …	… divide the numerator by the denominator.
To rewrite a percentage as a fraction, …	… multiply or divide by 100.
To convert between decimals and percentages, …	… use the place value of the digits to express the decimal as a fraction.

   b. Choose one phrase from the list on the left and complete it using a different strategy than what you matched it with above. Your strategy can be different from those listed above.
   Answers will vary. To rewrite a fraction as a percentage, divide the numerator by the denominator and multiply the quotient by 100.

Notice that there are multiple strategies students could use to convert from one form of a number to another. Students could draw more than one arrow to or from each phrase.

1.1.3 Guided Instruction: Repeating Decimals

Component Overview: *Students are introduced to repeating decimals. Consider a discussion of the vocabulary terms "repeating" and "terminating" and what they might look like. Students can use scientific calculators in this section.*

1.1.3 Guided Instruction: Repeating Decimals

A ***rational number*** is a real number that can be expressed as a ratio of two integers.

Rational numbers can be written as a fraction, $\frac{a}{b}$, where a and b are integers and $b \neq 0$, or as a decimal. In decimal form, rational numbers either terminate or repeat.

1. Consider the fractions in the table shown.

   a. Rewrite each fraction in its equivalent decimal form. Then, indicate whether each decimal is a terminating or repeating decimal. If a decimal repeats digits in a pattern, use an ellipse (…) to indicate that the pattern continues.

Fraction Form	Decimal Form	Terminating or Repeating?
$\frac{1}{3}$	0.33 …	○ terminating ○ repeating
$\frac{3}{8}$	0.375	○ terminating ○ repeating
$\frac{13}{6}$	2.166 …	○ terminating ○ repeating
$\frac{38}{9}$	4.22 …	○ terminating ○ repeating
$\frac{17}{5}$	3.4	○ terminating ○ repeating

Students can use calculators for this, which provides an opportunity to observe how to recognize repeating decimals using technology.
If you choose to have students do long division, that will lengthen the lesson.

What does it mean for a decimal to terminate? To repeat? What makes a repeating decimal?

b. Circle the fractions in the table that are greater than 1. Then, rewrite each as a mixed number.

$2\frac{1}{6}; 4\frac{2}{9}; 3\frac{2}{5}$

c. Based on your work in Part A, explain whether it is possible for a repeating decimal to ever terminate.
Answers will vary. It is not possible because the remainder in the dividend always repeats.

Prompt students to use an example to answer question 1c and use mathematical evidence in their explanation.

d. Discuss with your partner any patterns you notice about the denominators of fractions whose decimal equivalent is repeating.
Answers will vary. I notice that the denominators are multiples of three. I notice that the denominators are not factors of the numbers for decimal place values, such as 10 or 100.

To extend students' thinking, consider prompting students to see if they can come up with a conjecture for what types of fractions are repeating decimals vs. fractions that are terminating decimals. They will have an opportunity to test and revise their conjectures through the remainder of this lesson and Lesson 2.

A number with a decimal that never ends but has a pattern that repeats infinitely is called a **repeating decimal.**

Repeating decimals are written with a bar over the digits that repeat. For example, the fraction $\frac{13}{6}$ is expressed as $2.1\overline{6}$ when written as a decimal because the 6 is the digit that repeats infinitely.

2. Isabel divided 85 by 99 to rewrite $\frac{85}{99}$ in decimal form. She found the resulting pattern $0.858585\ldots$
Write $\frac{85}{99}$ in its equivalent decimal form.
$0.\overline{85}$

1.1.4 Exploration: Patterns in Repeating Decimals

Component Overview: *Students use scientific calculators to find the decimal equivalents of fractions. Then they use the relationships observed to make predictions about other decimal equivalents of fractions before verifying with their calculator.*

1.1.4 Exploration: Patterns in Repeating Decimals

1. Work with your partner to complete the following.

   a. Use a scientific calculator to find the decimal equivalent of each fraction.

Fraction	Decimal Equivalent
$\frac{1}{9}$	$0.\overline{1}$
$\frac{2}{9}$	$0.\overline{2}$
$\frac{3}{9}$	$0.\overline{3}$
$\frac{4}{9}$	$0.\overline{4}$
$\frac{5}{9}$	$0.\overline{5}$
$\frac{6}{9}$	$0.\overline{6}$

   b. Describe the relationship observed between fractions with a denominator of 9 and their decimal equivalents.
   Answers will vary. The relationship observed is that the numerator is the same as the digit that repeats in the decimal equivalent.

   c. Without performing any calculations, predict the decimal equivalent for each fraction in the table.

Fraction	Decimal Equivalent
$\frac{7}{9}$	$0.\overline{7}$
$3\frac{2}{9}$	$3.\overline{2}$
$17\frac{8}{9}$	$17.\overline{8}$

Students will need scientific calculators to complete this exploration. Consider using the Desmos scientific calculator.

Notice and Wonder

- *What do you notice about the repeating decimal equivalents when the denominator is 9?*
- *How could you use this pattern to apply to other fractions with a denominator of 9?*
- *How confident are you in your predictions?*

d. Verify your predictions using your calculator.
$0.\overline{7}$; $3.\overline{2}$; $17.\overline{8}$

2. Discuss with your partner how the values in the previous question could be used to determine the decimal equivalents for $\frac{1}{3}$ and $\frac{2}{3}$.
Answers will vary. To determine the equivalents, multiply the original fractions by $\frac{3}{3}$.

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.
Listen as students discuss their thinking in response to this prompt to determine students' thinking strategies to highlight or misconceptions to address.

a. Complete the equations.

$$\frac{1}{3} = \frac{\boxed{3}}{9}$$

$$\frac{2}{3} = \frac{\boxed{6}}{9}$$

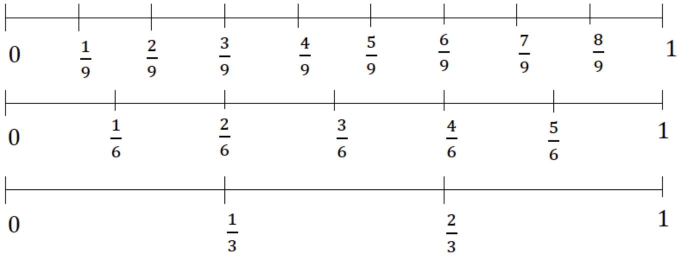
b. Complete the table.

Fraction	Decimal Equivalent
$\frac{1}{3}$	$0.\overline{3}$
$\frac{2}{3}$	$0.\overline{6}$
$\frac{7}{3} = \boxed{2}\frac{\boxed{1}}{3}$	$2.\overline{3}$

While students are working, notice if students are forgetting to put the bar over the digit in the repeating decimal. Consider a reminder that adding this bar is how repeating decimals are notated.

1.1.4 Exploration: Patterns in Repeating Decimals (continued)

3. Three number lines are shown.



a. Explain what the decimal equivalent of $\frac{2}{6}$ is using the number lines.
Answers will vary. $\frac{2}{6}$ is located in the same place as the $\frac{3}{9}$ and $\frac{1}{3}$, so the three fractions are equivalent. Therefore, $\frac{2}{6}$ is $0.\overline{3}$ when expressed as a decimal.

b. The decimal equivalent of $\frac{1}{6}$ is also a repeating decimal. Work with your partner to predict the decimal equivalent of $\frac{1}{6}$ based on the value of $\frac{2}{6}$.
Answers will vary. We predict the equivalent is half of the equivalent for $\frac{2}{6}$, which is $0.1\overline{6}$

c. Use a calculator to find the decimal equivalent of $\frac{1}{6}$. Write the value in decimal form.
$0.1\overline{6}$

Notice that some calculators may round the last digit shown on the screen. This may confuse some students. Consider a discussion on how to recognize a repeating decimal on a calculator.

d. Explain whether the decimal equivalent of $\frac{3}{6}$ is a repeating or terminating decimal.
Answers will vary. It is terminating because 3 is exactly half of 6, and the equivalent decimal of $\frac{1}{2}$ is terminating.

1.1.5 Bring It Together: Patterns in Repeating Decimals

Component Overview: *Students make predictions about which fractions have decimal equivalents that are repeating based on observed patterns from the Exploration. They then verify the accuracy of their predictions and make conjectures about which fractions have decimal expansions that are repeating. Students will have an opportunity to verify and refine their conjectures in Lesson 2.*

1.1.5 Bring It Together: Patterns in Repeating Decimals

In the Exploration, you observed most fractions with denominators 3, 6, or 9 have a decimal equivalent that repeats.

1. Based on your observations, circle the fractions below that you predict also have repeating decimal equivalents.
   Answers will vary.
   $\frac{1}{11}$ $\frac{1}{12}$ $\frac{3}{12}$ $\frac{6}{15}$ $\frac{2}{15}$ $\frac{12}{99}$

2. Use a scientific calculator to verify your predictions. Write the decimal equivalent for each fraction in the table.

Fraction	Decimal Equivalent
$\frac{1}{11}$	$0.\overline{09}$
$\frac{1}{12}$	$0.08\overline{3}$
$\frac{3}{12}$	0.25
$\frac{6}{15}$	0.4
$\frac{2}{15}$	$0.1\overline{3}$
$\frac{12}{99}$	$0.\overline{12}$

3. Make a conjecture about which fractions have decimal expansions that are repeating.
   Answers will vary. Fractions that can be simplified to have a 3, 9, or 11 in the denominator are repeating decimals.

Consider asking students the following:

- *What patterns do you notice?*
- *What do fractions of the repeating decimals have in common?*
- *What do the fractions of the terminating decimals have in common?*

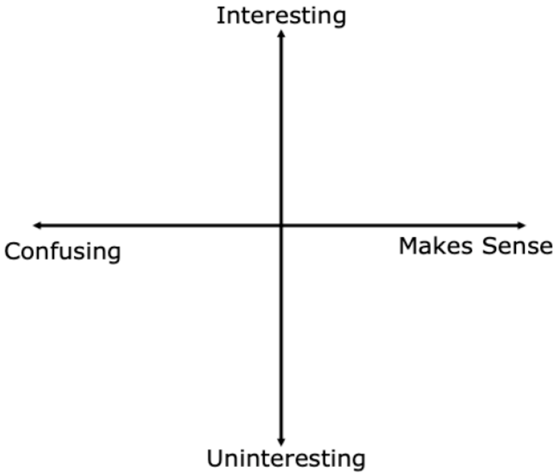
Think-Pair-Share
It is not expected that students come up with a rule that always works. However, students' conjectures will reveal helpful information about what patterns they are noticing in these relationships.
Consider implementing a Think-Pair-Share here and having students share what they notice, then talking about these observations as a class.

1.1.6 Wrap-Up: Reflection

Component Overview: *Students complete a self-reflection based on what they learned in this lesson.*

1.1.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate where you are on today's learning targets.
   Answers will vary.



2. Write one question you still have about what you learned today.
   Answers will vary.

Consider going through the questions that students write and choosing the top few questions that occur the most to review at the beginning of Lesson 2.